Perceptual Organization (III)

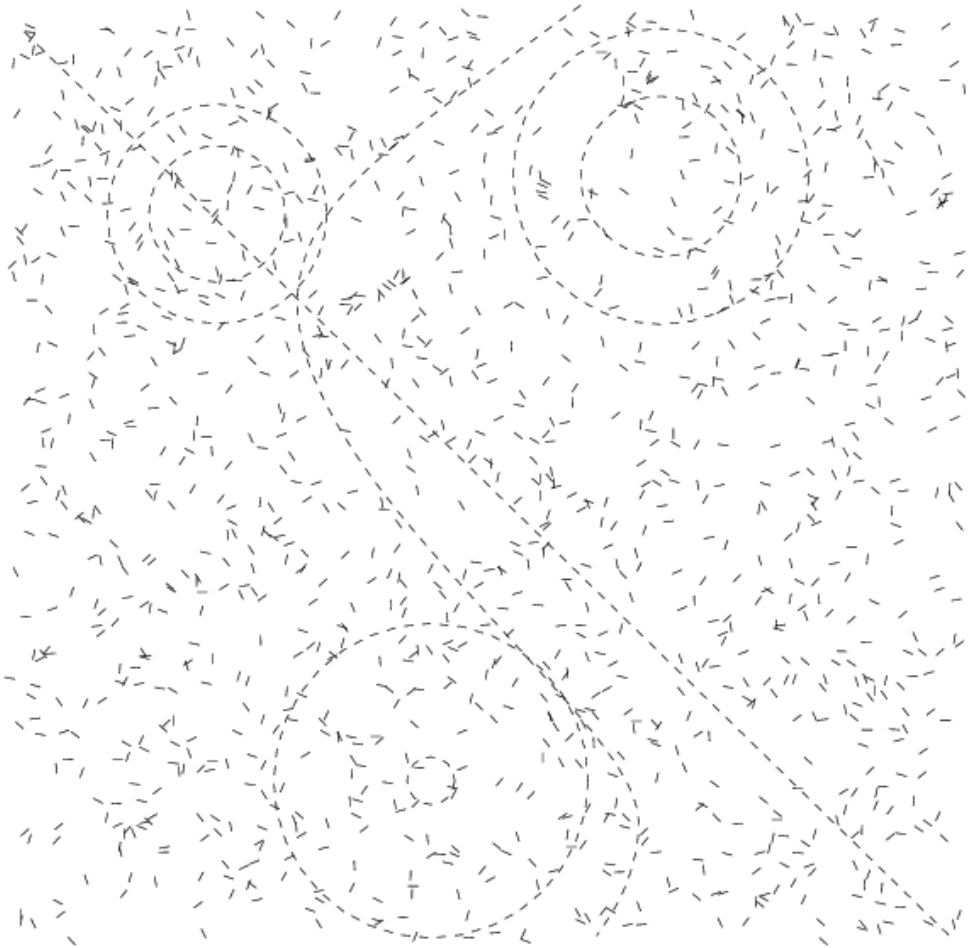
Introduction to Computational and Biological Vision

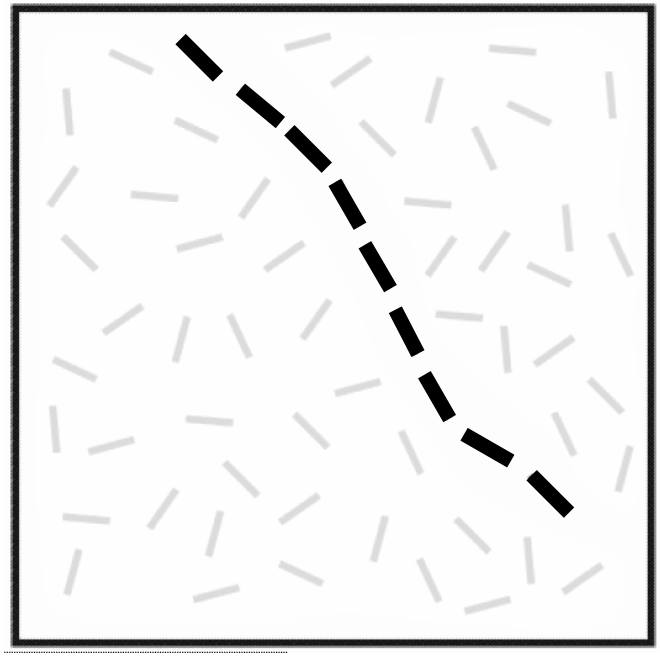
CS 202-1-5261

Computer Science Department, BGU

Ohad Ben-Shahar

Curve Inference via Good Continuation

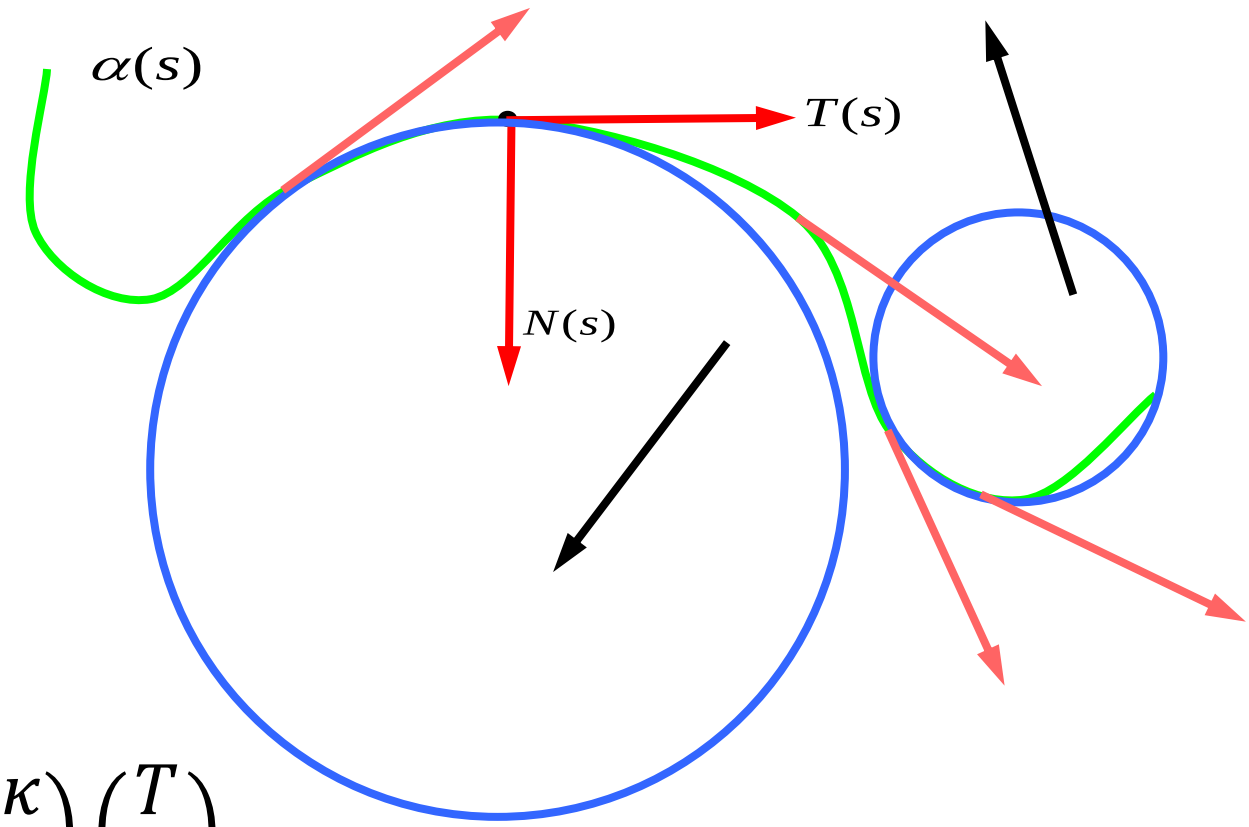




Curve Inference via Good Continuation

Curve Inference via Good Continuation

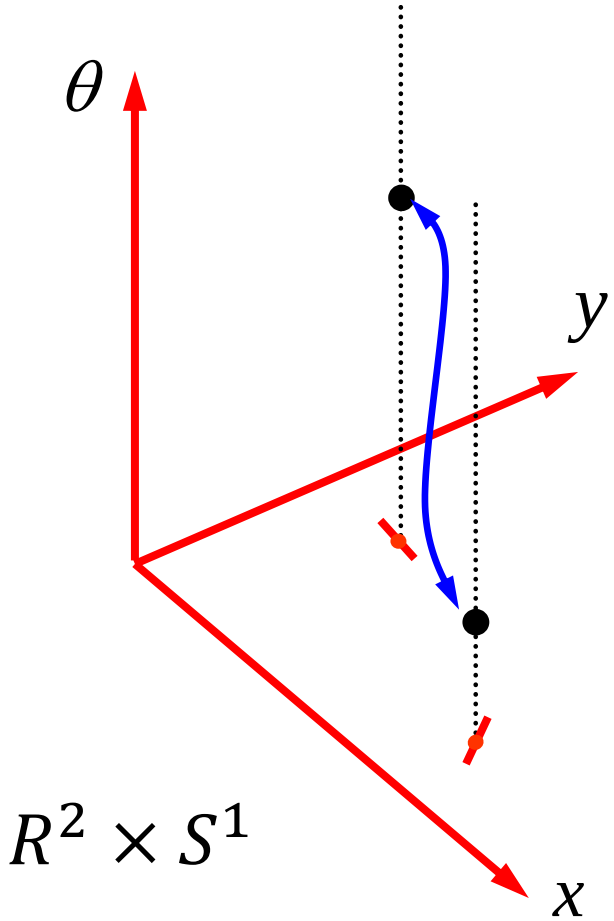
Good continuation of edge elements

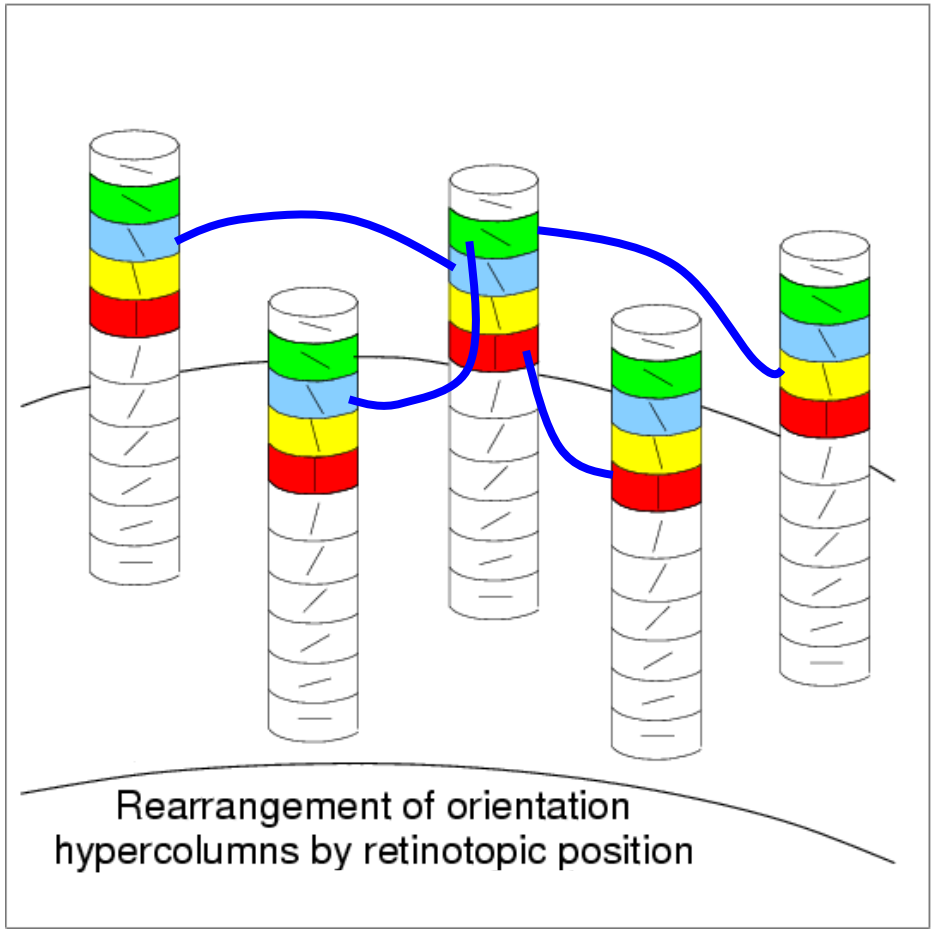


$$\begin{pmatrix} T' \\ N' \end{pmatrix} = \begin{pmatrix} 0 & \kappa \\ -\kappa & 0 \end{pmatrix} \begin{pmatrix} T \\ N \end{pmatrix}$$

Curve Inference via Good Continuation

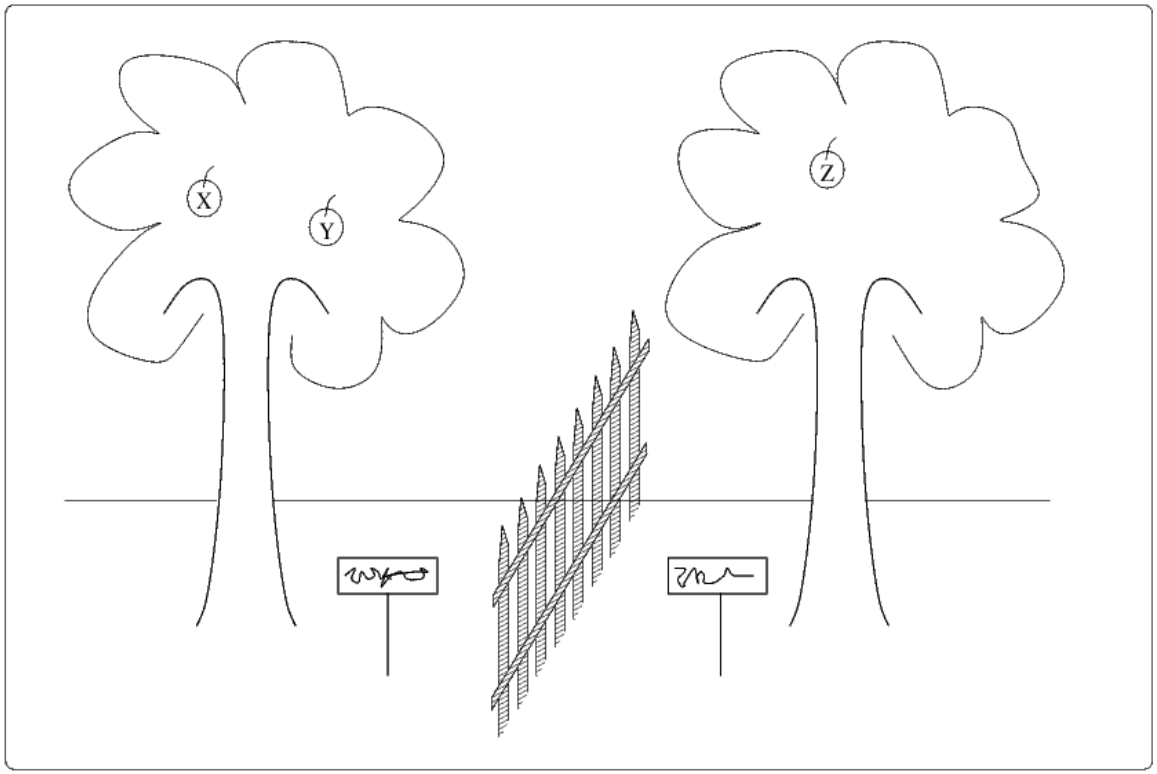
Reminder – tangent interactions and the primary visual cortex





Curve Inference via Good Continuation

Relaxation labeling



Contextual constraints

$$X = Y$$

$$X \neq Z$$

$$Y \neq Z$$

$$\downarrow$$

$$X = apple$$

$$Y = apple$$

$$Z = orange$$

Measurements

$P_X(\text{Apple}) = 100\%$ $P_Y(\text{Apple}) = 50\%$ $P_Z(\text{Apple}) = 50\%$
$P_X(\text{Orange}) = 0\%$ $P_Y(\text{Orange}) = 50\%$ $P_Z(\text{Orange}) = 50\%$

Curve Inference via Good Continuation

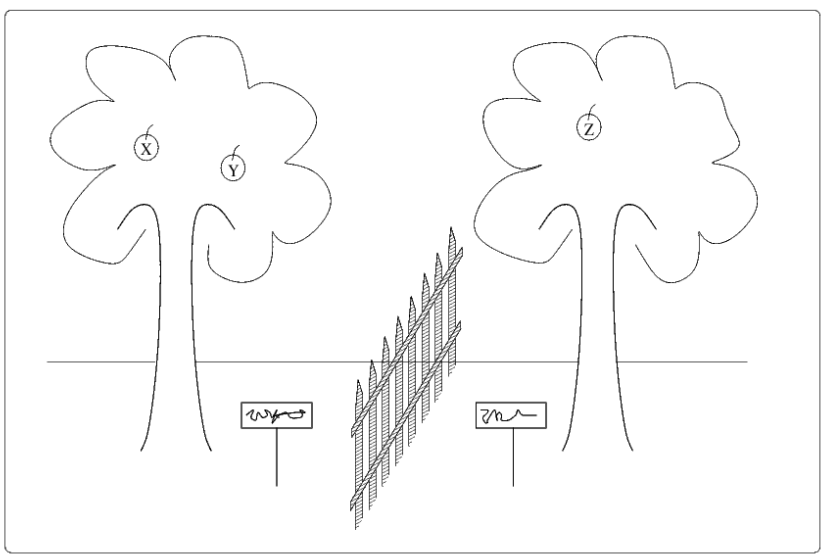
Relaxation labeling

$B = \{b_1, \ldots, b_n\}$	set of **objects** to be labeled
$\Lambda = \{1, 2, \ldots, m\}$	set of possible **labels**
$p_i^0(\lambda)$	the measured **confidence** that b_i should be labeled λ. $p_i^0(\lambda) \geq 0$ $\sum_{\lambda=1}^{m} p_i^0(\lambda) = 1 \quad \forall i$

Curve Inference via Good Continuation

Relaxation labeling

$r_{ij}(\lambda, \mu)$ $\Leftrightarrow$ the strength of **compatibility** between the hypotheses "b_i has label λ" and "b_j has label μ"



$r_{XY}(\lambda,\mu)$	A	O
A	1	0
O	0	1

$r_{YX}(\lambda,\mu)$	A	O
A	1	0
O	0	1

$r_{XX}(\lambda,\mu)$	A	O
A	1	0
O	0	1

$r_{XZ}(\lambda,\mu)$	A	O
A	0	1
O	1	0

$r_{ZX}(\lambda,\mu)$	A	O
A	0	1
O	1	0

$r_{YY}(\lambda,\mu)$	A	O
A	1	0
O	0	1

$r_{YZ}(\lambda,\mu)$	A	O
A	0	1
O	1	0

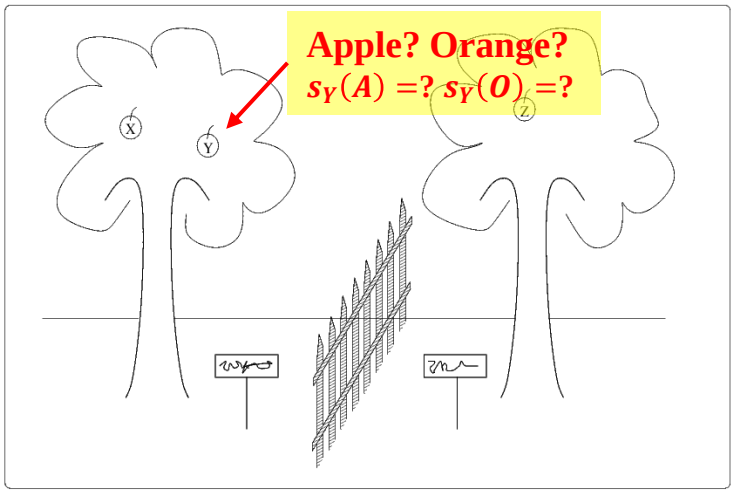
$r_{ZY}(\lambda,\mu)$	A	O
A	0	1
O	1	0

$r_{ZZ}(\lambda,\mu)$	A	O
A	1	0
O	0	1

Curve Inference via Good Continuation

Relaxation labeling

Support for label λ at b_i: $$s_i(\lambda) = s(i, \lambda; \bar{r}, \bar{p}) = \sum_{j=1}^{n} \sum_{\mu=1}^{m} r_{ij}(\lambda, \mu) p_j(\mu)$$



Measurements

$P_X(\text{Apple}) = 100\%$
$P_X(\text{Orange}) = 0\%$

$P_Y(\text{Apple}) = 50\%$
$P_Y(\text{Orange}) = 50\%$

$P_Z(\text{Apple}) = 50\%$
$P_Z(\text{Orange}) = 50\%$

$r_{YX}(\lambda, \mu)$	A	O
A	1	0
O	0	1

$r_{YY}(\lambda, \mu)$	A	O
A	1	0
O	0	1

$r_{YZ}(\lambda, \mu)$	A	O
A	0	1
O	1	0

$$s_Y(A) = r_{YX}(A,A)p_X(A) + r_{YX}(A,O)p_X(O) + r_{YY}(A,A)p_Y(A) + r_{YY}(A,O)p_Y(O) + r_{YZ}(A,A)p_Z(A) + r_{YZ}(A,O)p_Z(O)$$

$$s_Y(A) = 1 \cdot 1 + 0 \cdot 0 + 1 \cdot 0.5 + 0 \cdot 0.5 + 0 \cdot 0.5 + 1 \cdot 0.5 = 2$$

$$s_Y(O) = r_{YX}(O,A)p_X(A) + r_{YX}(O,O)p_X(O) + r_{YY}(O,A)p_Y(A) + r_{YY}(O,O)p_Y(O) + r_{YZ}(O,A)p_Z(A) + r_{YZ}(O,O)p_Z(O)$$

$$s_Y(O) = 0 \cdot 1 + 1 \cdot 0 + 0 \cdot 0.5 + 1 \cdot 0.5 + 1 \cdot 0.5 + 0 \cdot 0.5 = 1$$

Curve Inference via Good Continuation

Relaxation labeling

Space of all possible label assignments:

$$\mathbb{K} = \{\bar{p} = (\bar{p}_1, \ldots, \bar{p}_n) : \quad \bar{p}_i = (p_i(1), \ldots, p_i(m)) \in \mathbb{R}^m,$$

$$0 \leq p_i(\lambda) \leq 1 \quad \forall i, \lambda \ ,$$

$$\textstyle\sum_{\lambda=1}^{m} p_i(\lambda) = 1 \quad \forall i\}$$

What would be desired label assignments?

When should we consider a labeling to be consistent?

Curve Inference via Good Continuation

Relaxation labeling

Space of unambiguous labeling

$$\mathbb{K}^* = \{\bar{p} = (\bar{p}_1, \ldots, \bar{p}_n) : \quad \bar{p}_i = (p_i(1), \ldots, p_i(m)) \in \mathbb{R}^m,$$

$$p_i(\lambda) \in \{0, 1\} \quad \forall i, \lambda \ ,$$

$$\textstyle\sum_{\lambda=1}^{m} p_i(\lambda) = 1 \quad \forall i\}$$

Labeling $\bar{p}$ is consistent in $\mathbb{K}^*$ if

$$\left.\begin{array}{l} s_1(\lambda_1; \bar{p}) \geq s_1(\bar{\lambda}; \bar{p}) \quad \forall \bar{\lambda} \\ \vdots \\ s_n(\lambda_n; \bar{p}) \geq s_n(\bar{\lambda}; \bar{p}) \quad \forall \bar{\lambda} \end{array}\right\} \text{Simultaneously}$$

Curve Inference via Good Continuation

$s_1\,(\lambda_1;\bar{p}) \geq s_1\,(\bar{\lambda};\bar{p})$ **Relaxation labeling**

$$\overset{1}{p_1(\lambda_1)}s_1\,(\lambda_1;\bar{p}) + \cdots + \overset{0}{p_1(\bar{\lambda})}s_1\,(\bar{\lambda};\bar{p}) + \cdots + \overset{0}{p_1(\lambda_m)}s_1\,(\lambda_m;\bar{p})$$

$$\geq$$

$$\overset{0}{q_1(\lambda_1)}s_1\,(\lambda_1;\bar{p}) + \cdots + \overset{1}{q_1(\bar{\lambda})}s_1\,(\bar{\lambda};\bar{p}) + \cdots + \overset{0}{q_1(\lambda_m)}s_1\,(\lambda_m;\bar{p})$$

Different permutation of {0,1} values

So instead of $s_i\,(\lambda;\bar{p}) \geq s_i\,(\bar{\lambda};\bar{p})\ \ \forall i,\bar{\lambda}$
we define consistency by

<u>Definition</u>: $\bar{p}$ is consistent in $\mathbb{K}^*$ if

$$\sum_{\lambda=1}^{m} p_i(\lambda)s_i(\lambda;\bar{p}) \geq \sum_{\lambda=1}^{m} q_i(\lambda)s_i(\lambda;\bar{p}) \quad \forall i$$

Curve Inference via Good Continuation

Definition: $\bar{p}$ is consistent in $\mathbb{K}^*$ if

$$\sum_{\lambda=1}^{m} p_i(\lambda) s_i(\lambda; \bar{p}) \geq \sum_{\lambda=1}^{m} q_i(\lambda) s_i(\lambda; \bar{p}) \quad \forall i$$

Define consistency for *ambiguous* labeling in the same way and apply for the space of weighted labeling assignments

A labeling $\bar{p} = \{p_i(\lambda)\}$ is said to be **consistent** if

$$\sum_{\lambda=1}^{m} p_i(\lambda) s_i(\lambda, \bar{p}) \geq \sum_{\lambda=1}^{m} q_i(\lambda) s_i(\lambda, \bar{p}) \quad \forall \ \bar{q} \in K$$

for all $i = 1..n$ simultaneously.

Curve Inference via Good Continuation

Relaxation labeling

The **average local consistency** of the assignment

$$A(\bar{p}) = \sum_i \sum_\lambda p_i(\lambda) s_i(\lambda)$$

The original "ad-hoc" **update rule** [Rosenfeld *et. al.* 1976]:

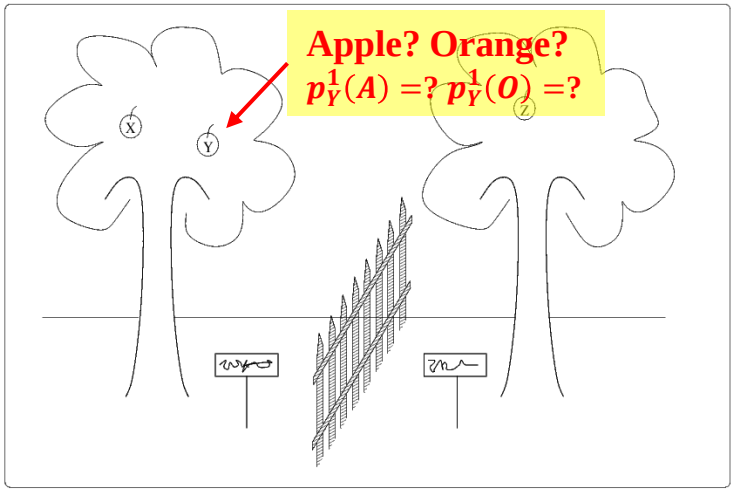
$$p_i^{k+1}(\lambda) = \frac{p_i^k(\lambda) s_i^k(\lambda)}{\sum_{\mu=1}^m p_i^k(\mu) s_i^k(\mu)}$$

Fundamental result of RL theory [Hammel &Zucker 83]:

Under certain conditions on $r_{ij}(\lambda, \mu)$ the relaxation labeling process will converge to a consistent labeling while maximizing the average local consistency.

Curve Inference via Good Continuation

Relaxation labeling



Measurements

$P_X(\text{Apple}) = 100\%$ $P_Y(\text{Apple}) = 50\%$ $P_Z(\text{Apple}) = 50\%$
$P_X(\text{Orange}) = 0\%$ $P_Y(\text{Orange}) = 50\%$ $P_Z(\text{Orange}) = 50\%$

$$s_Y^0(A) = 2$$

$$s_Y^0(O) = 1$$

$$p_i^{k+1}(\lambda) = \frac{p_i^k(\lambda) s_i^k(\lambda)}{\sum_{\mu=1}^{m} p_i^k(\mu) s_i^k(\mu)}$$

$$p_Y^1(A) = \frac{p_Y^0(A) \cdot s_Y^0(A)}{p_Y^0(A) \cdot s_Y^0(A) + p_Y^0(O) \cdot s_Y^0(O)} = \frac{0.5 \cdot 2}{0.5 \cdot 2 + 0.5 \cdot 1} = \frac{2}{3} = 66\%$$

$$p_Y^1(O) = \frac{p_Y^0(O) \cdot s_Y^0(O)}{p_Y^0(A) \cdot s_Y^0(A) + p_Y^0(O) \cdot s_Y^0(O)} = \frac{0.5 \cdot 1}{0.5 \cdot 2 + 0.5 \cdot 1} = \frac{1}{3} = 33\%$$

Curve Inference via Good Continuation

Relaxation labeling for curve inference via good continuation

Nodes:

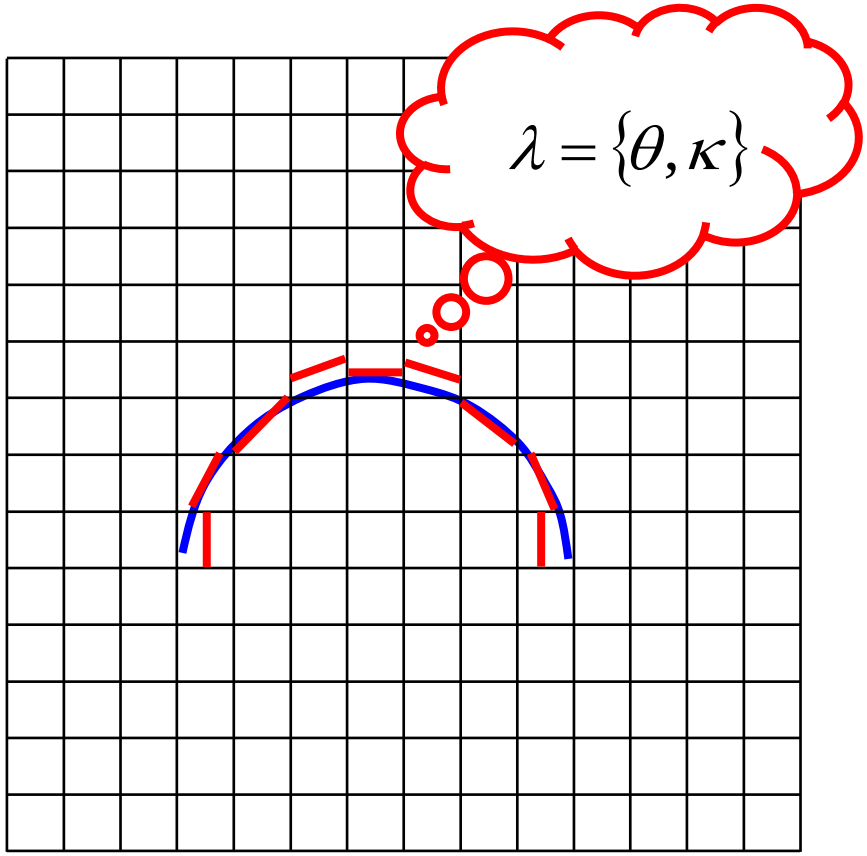
Image pixels (x,y)

Labels:

Orientation+curvature

Compatibilities:

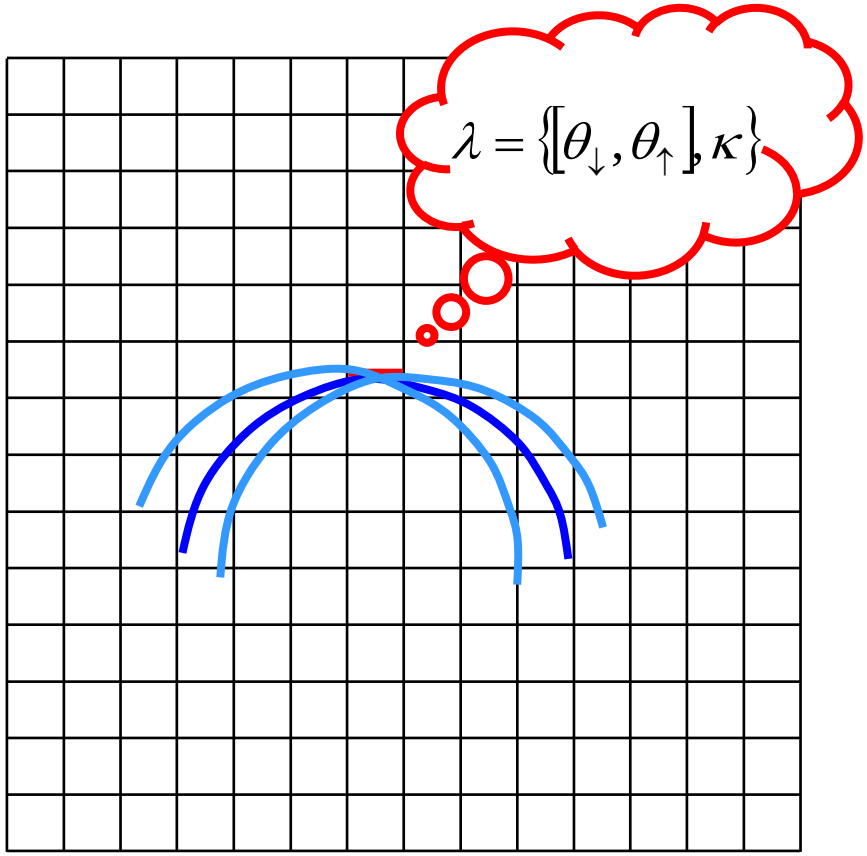
Based on cocircularity



Curve Inference via Good Continuation

Relaxation labeling for curve inference via good continuation

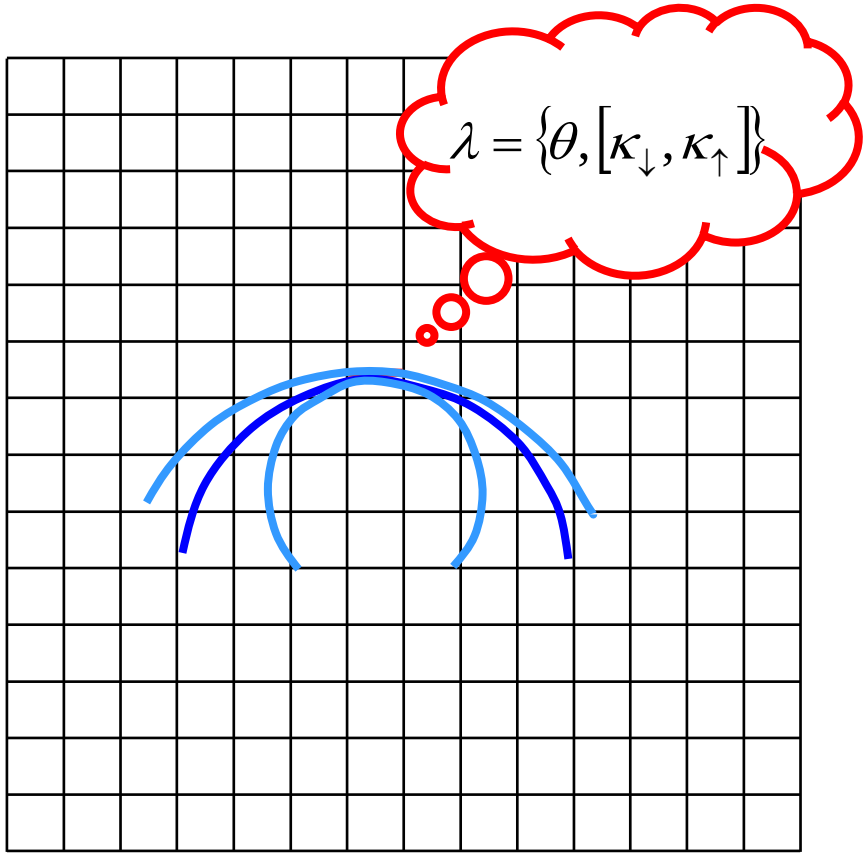
Label quantization: Orientation



Curve Inference via Good Continuation

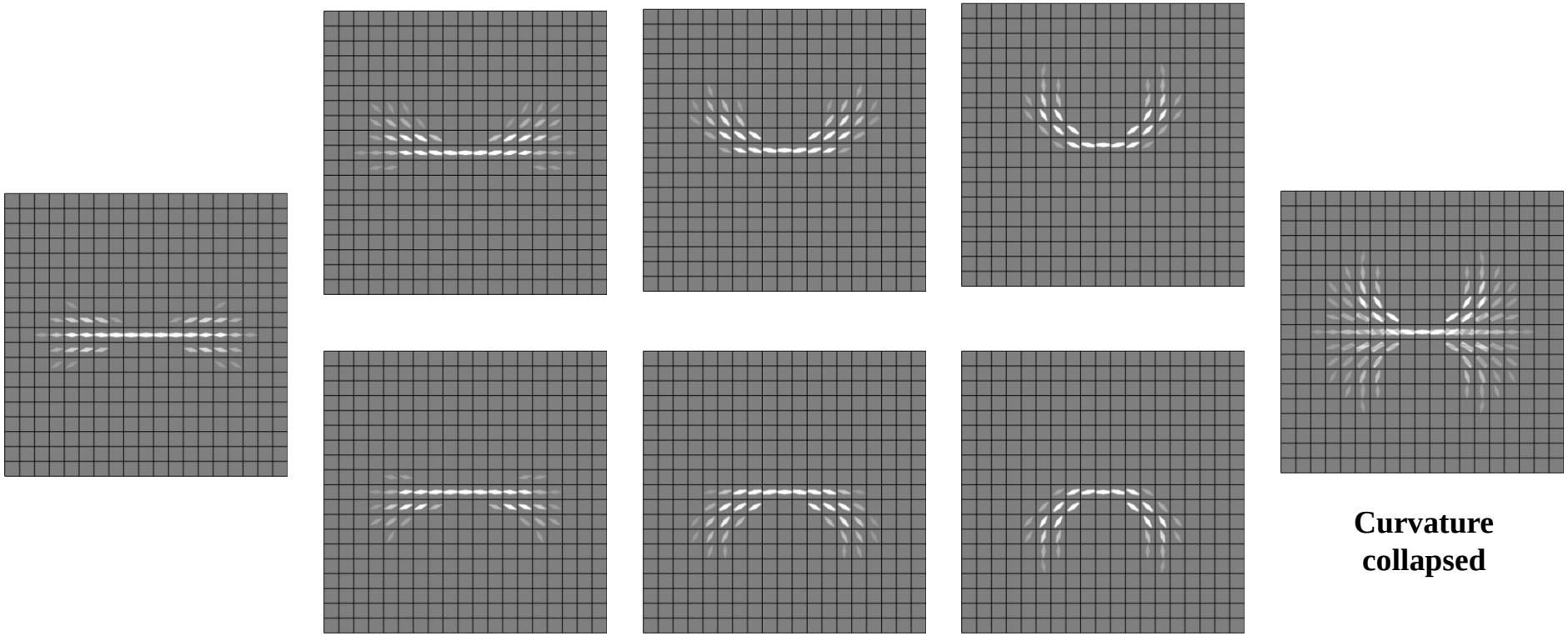
Relaxation labeling for curve inference via good continuation

Label quantization: Curvature



Curve Inference via Good Continuation

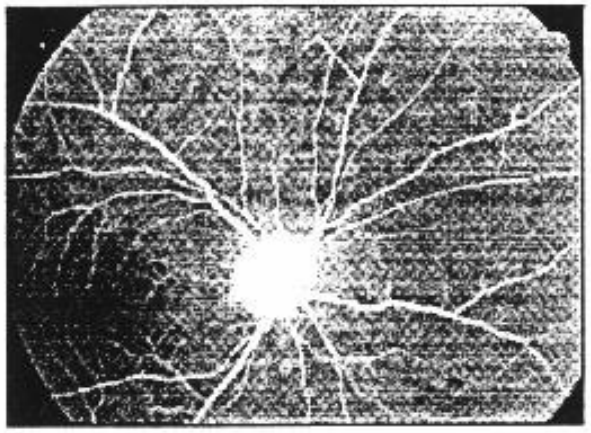
Relaxation labeling for curve inference via good continuation



Cocircular compatibility fields

Curve Inference via Good Continuation

Relaxation labeling for curve inference via good continuation

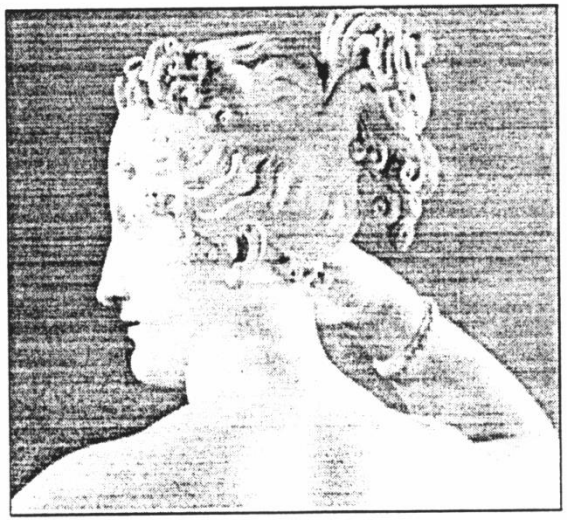


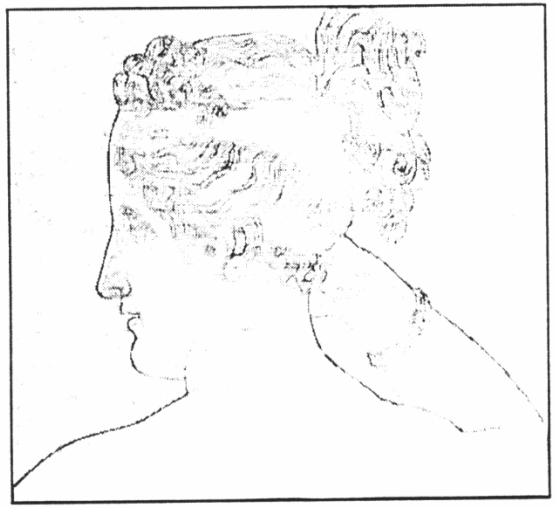
Curve Inference via Good Continuation

Relaxation labeling for curve inference via good continuation

Curve Inference via Good Continuation

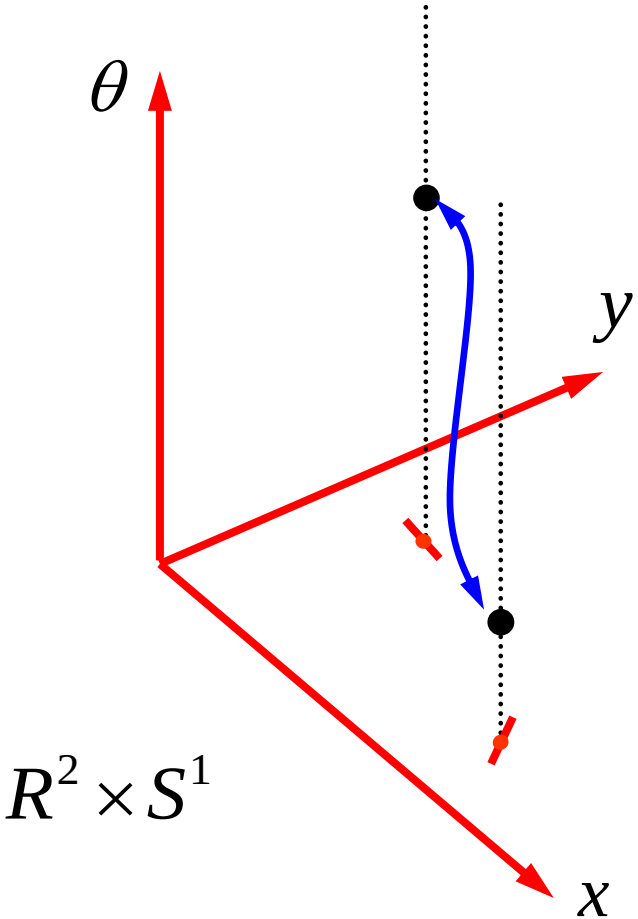
Relaxation labeling for curve inference via good continuation

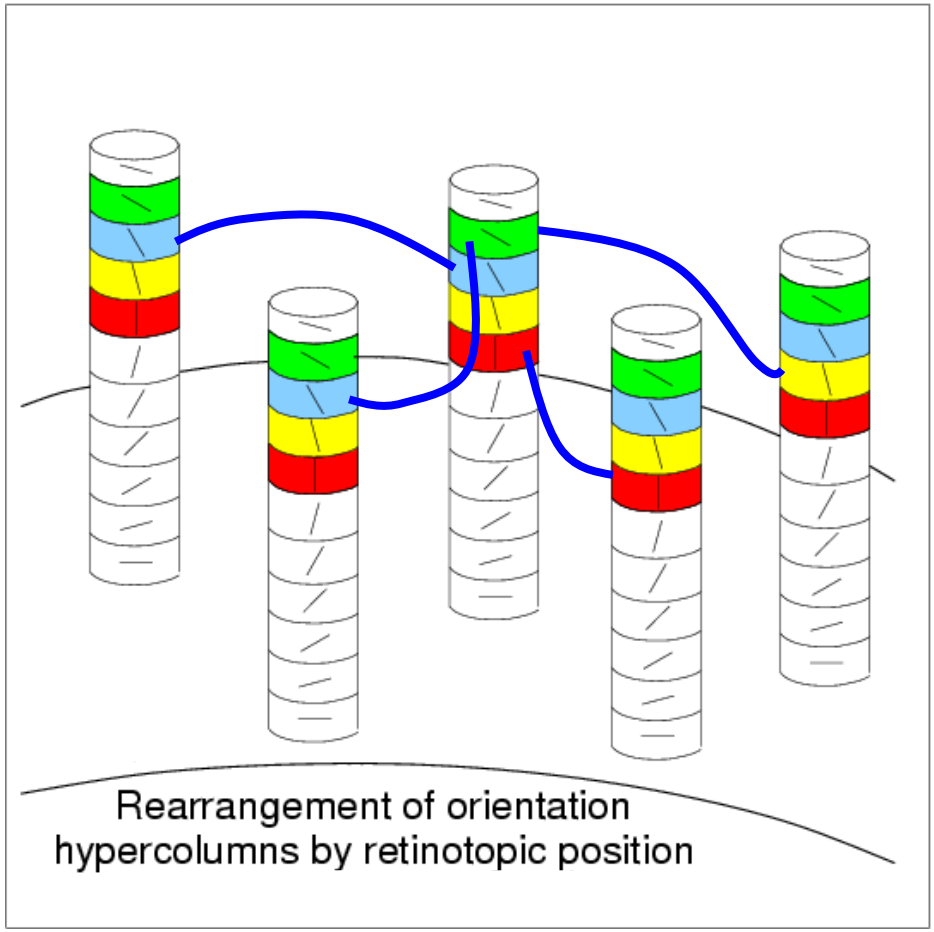


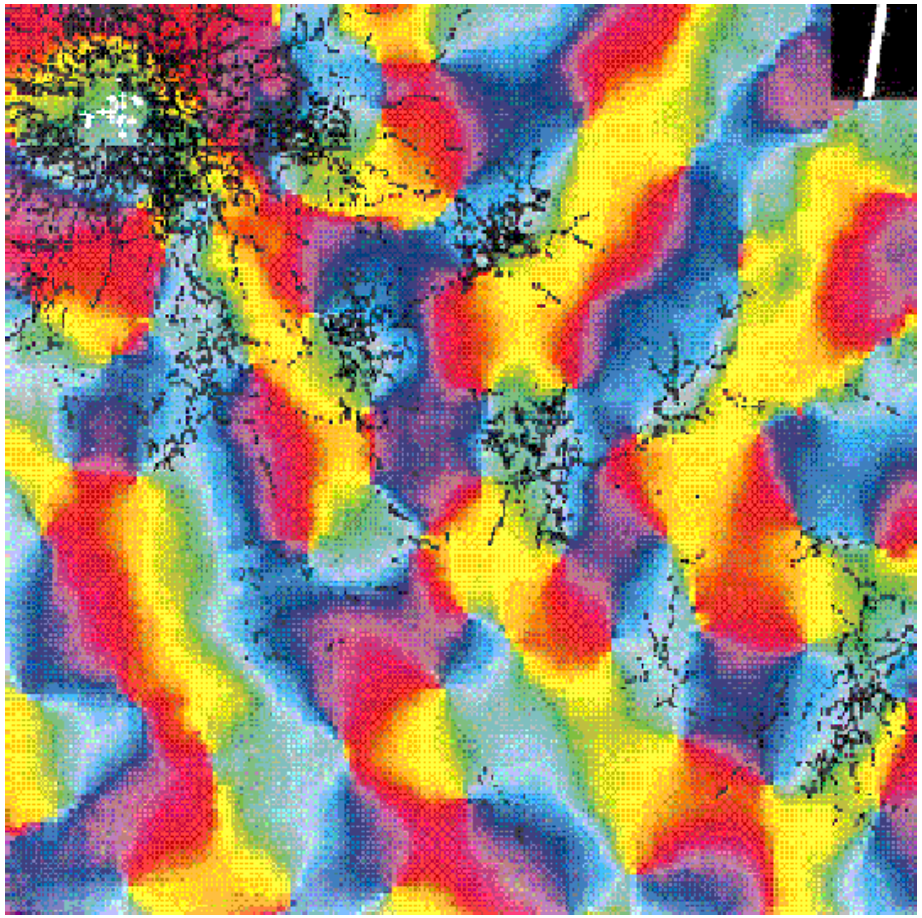


Curve Inference via Good Continuation

Is it really biologically plausible?





Curve Inference via Good Continuation

Is it really biologically plausible?